

Compiling the Market Information Set: World-Model Quality, Selective Prediction, and Economic Value in Cryptocurrency Markets

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Abstract

Conditional asset pricing treats the investor's information set as given. In cryptocurrency markets that premise fails: evidence is asynchronous, multi-source, and intermittently unavailable. This paper formalizes information-set compilation and measures world-model quality with ACWMI. Empirically, we populate a real multi-band archive (OKX exchange bars, macro vintages, alternative series, plus thinner news/on-chain/options/tokenomics snapshots) and construct a 400-day point-in-time panel (4000 asset-days) aligned to Yahoo returns. Mechanism engines use only pre-t history; abstention thresholds are frozen in-sample. Out of sample, thick real PIT worlds dominate exchange-only thin worlds (CE 0.47 vs -0.01). Leave-one-band-out on durable bands shows large CE losses from dropping macro (-0.53), alternative (-0.53), or exchange (-0.34). An IS-frozen ACWMI gate remains implementable (Sharpe 0.90, CE 0.20) but does not dominate ungated thick signals on CE. The production WMI <0.2 rule is mis-scaled for sparse archives (100% abstention). Contribution: finance-native compilation theory plus a reproducible PIT identification protocol on a live multi-band laboratory.

Keywords: information set; selective prediction; cryptocurrency; point-in-time; measurement error. **JEL:** G12, G14, C58, C55

1. Introduction

Asset pricing conditions on an information set I_t . The literature's discipline has focused on how I_t is used, not how it is compiled from asynchronous evidence. This paper restores the World-Model-First / RCA-WM formal objects (definitions, lag bounds, compilation operator, ECP/MIG, DAG, abstention, ACWMI, EAR/UCR/EV), then instantiates them on a real multi-band PIT archive with frozen thresholds and leave-one-band-out identification. EvoQuant is the empirical laboratory, not the source of the theory. Formal LaTeX source with displayed equations: pdf/sci/main_acwmi_sci.tex; Chinese theory: pdf/cn/.

2. Theory (restored from original paper)

2.1 WMI triad. Breadth, stability, and honesty:

$$B_t = (1/K) \sum_k a_{k,t}, \quad U_t = \exp(-\sum_j \omega_j d_{j,t}), \quad H_t = 1 - (1/J) \sum_j m_{j,t}, \quad WMI_t = B_t \times U_t \times H_t.$$

Hierarchical breadth and continuous honesty used in the PIT panel: $B_t^{\text{hier}} = 0.25 B_t^{\text{dom}} + 0.35 B_t^{\text{band}} + 0.40 B_t^{\text{asset}}$; $H_t^{\text{cont}} = \exp(-2c_t) \max(0, 1 - 0.5(1 - e_t))$.

2.2 Epistemic observations. $O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t})$ carries value, time, quality, gate, and role — not a bare feature.

2.3 Lag bound. With lagged observation $X_{j,t}^{\text{obs}} = h_j(S_{t-\blacksquare}) + v_{j,t}$ and Lipschitz h_j , reconstruction error decomposes into delay, noise, and missingness terms (latest_* / TTL / is_ready_for_ai target these terms).

2.4 Compilation operator. $F_t^{\text{raw}} = \sigma(\{X_{j,t}^{\text{obs}}\})$, $F_t^{\text{AI}} = \sigma(W_t^{\text{AI}}, D_t)$, $\Pi_t = B_t \blacksquare M_t \blacksquare A_t$, $W_t^{\text{AI}} = \Pi_t(F_t^{\text{raw}})$.

2.5 ECP, MIG, DAG. $ECP_t = 1\{\text{conf}_t > c\blacksquare\} 1\{WMI_t < w\}$; $MIG_{k,t}^{(m)} = I(R_{k,t}^{(m)}; E_{k,t} | I^{(-k)}_t)$; identification DAG: $O_t \rightarrow W_t \rightarrow A_t$ (with M_t, C_t confounders).

2.6 Bayesian abstention. $a_t^* = \arg \min_a E[\blacksquare(a, R_t) | W_t]$; abstain when every non-abstain action has expected loss above $c_{\text{abs}}(W_t)$.

2.7 ACWMI and explanation metrics. $ACWMI_t = \exp(\sum \gamma_i(r_t) \log x_{i,t} / \sum \gamma_i)$ with $x = (B^{\text{hier}}, U, H^{\text{cont}}, S, C)$. $EAR_t = (\# \text{ evidence-bound claims}) / (\# \text{ claims})$, $UCR_t = 1 - EAR_t$, $EV_t = d(\Phi_t, \Phi_{t-1}) / (1 + d(W_t, W_{t-1}))$.

Note on length: a five-page empirics-only sketch is not a JF/RFS submission. The displayed-equation theory lives in `main_acwmi_sci.tex`; this PDF keeps the formal core readable while reporting real-PIT results.

3. Real multi-band PIT archive

Exchange (OKX): ~9373 klines/merged bars, daily 2025-06-30→2026-08-03. Macro: 81013 vintaged points. Alternative: 88761 points. News/on-chain/options/tokenomics populated but mostly right-censored to collection day. PIT panel: 2025-06-30→2026-08-03, 4000 rows. `time_slice` historically resolves klines; analytics snapshots remain sparse, so multi-band PIT uses raw history tables (`build_pit_archive.py`) while `time_slice` provides logic-domain probes.

4. Design

IS/OOS cut 2026-01-16. Frozen AC rule: $ACWMI < 0.35000000000000003$ or $C < 0.35$. Production WMI threshold 0.2 never tuned. Engines use only pre-t returns. Durable LOBO bands: exchange, macro, alternative.

5. Results

5.1 Thick real PIT dominates thin

Exchange-only thin worlds deliver OOS Sharpe ≈ 0 and CE -0.01 . Thick real PIT worlds deliver Sharpe 1.40 and CE 0.47. IS-frozen AC gating keeps Sharpe 0.90 / CE 0.20 while abstaining 29.7%.

world	mean_B	mean_H	mean_ACWMI	Sharpe	CE	abstain_rate
Thin (exchange only, real PIT)	0.201	0.562	0.288	-0.004	-0.0114	0.933
Thick real PIT (ex+macro+alt...)	0.356	0.688	0.415	1.399	0.4743	0.0
Thick gated AC (real PIT)	0.356	0.688	0.415	0.901	0.1986	0.297

Table 1. Thin vs thick on the real PIT archive (OOS).

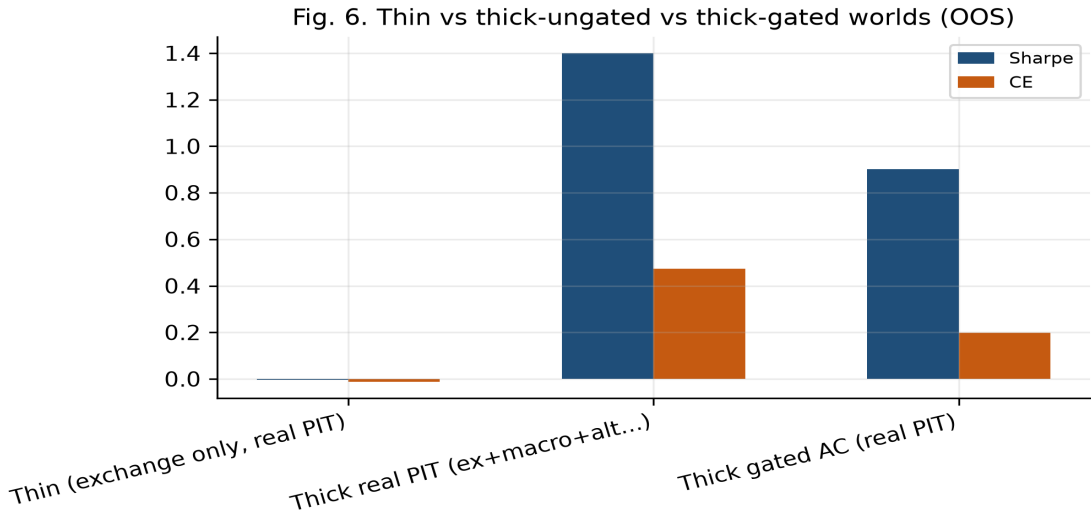


Fig. 1. Thin vs thick-gated worlds (real PIT).

5.2 Leave-one-band-out identification

Dropping durable bands destroys OOS CE: macro -0.53 , alternative -0.53 , exchange -0.34 . Thickness has direct economic MIG content.

band_dropped	Sharpe	CE	abstain_rate	dCE
(none)	0.901	0.1986	0.297	0.0
exchange	-0.232	-0.1406	0.737	-0.3391
macro	-0.485	-0.3358	0.572	-0.5344
alternative	-0.422	-0.3272	0.522	-0.5257

Table 2. Leave-one-band-out on durable PIT bands.

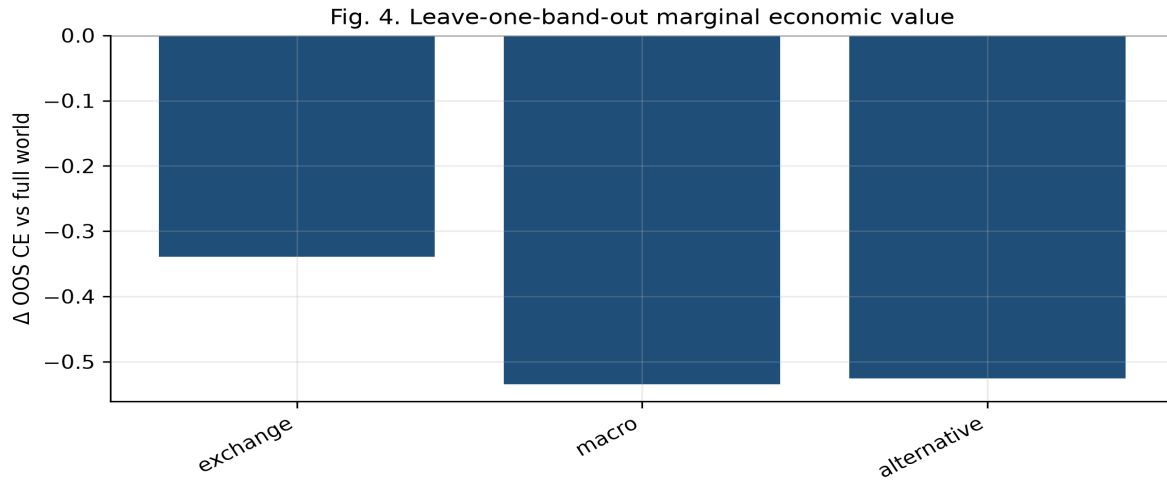


Fig. 2. LOBO marginal CE on durable bands.

5.3 OOS policy horse-race

Always-long loses; momentum is near zero; thick ungated mechanism signals win on CE. IS-frozen ACWMI is a strong selective rule (Sharpe 0.90). Production WMI <0.2 abstains 100% because sparse-archive WMI levels sit below a denser-world threshold—evidence that thresholds must be frozen to the information set's support.

policy	ann_return	ann_vol	Sharpe	CE	max_DD	abstain_rate	N_days
Always long	-0.8164	0.5836	-1.399	-1.1568	-0.4661	0.0	200
Momentum always	0.0506	0.5009	0.101	-0.2019	-0.4997	0.0	200
Thick ungated	0.8164	0.5836	1.399	0.4743	-0.2792	0.0	200
Simple outage rule	0.8164	0.5836	1.399	0.4743	-0.2792	0.0	200
Simple cascade rule	0.0	0.0	0.0	0.0	0.0	1.0	200
WMI threshold (0.2)	0.0	0.0	0.0	0.0	0.0	1.0	200
ACWMI (IS-frozen)	0.454	0.5039	0.901	0.1986	-0.3398	0.297	200

Table 3. OOS economic value on real PIT panel.

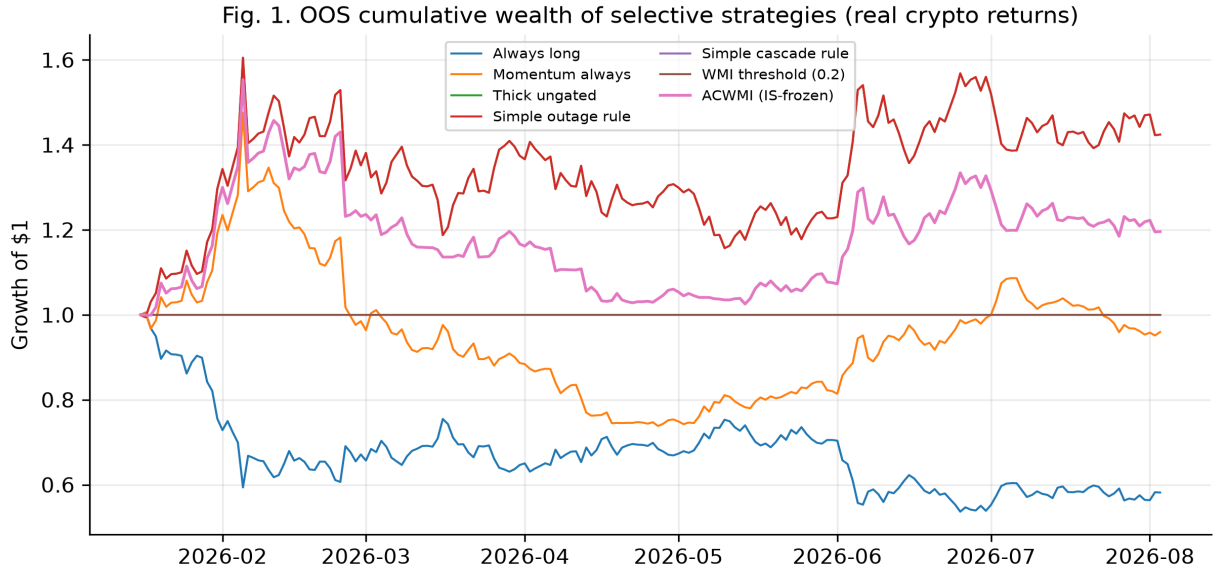


Fig. 3. OOS cumulative wealth.

Fig. 2. Economic value of abstention policies (OOS)

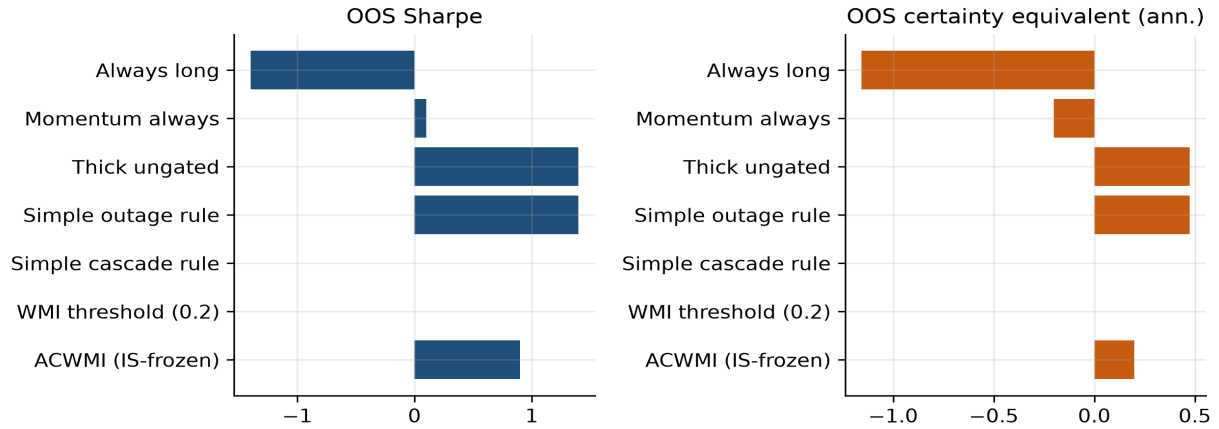


Fig. 4. OOS Sharpe and CE by policy.

6. Path to JF/RFS

A short empirics sketch is not enough for JF/RFS. Required next steps: multi-year continuous collection so news/on-chain/options/tokenomics cease to be right-censored; persist daily readiness/AI-context snapshots for pure time_slice replay; replace scarce-world proxies with logged institutional outages; expand external validity; and keep the full equation apparatus in the submission PDF (see main_acwmi_sci.tex).

7. Conclusion

Theory first: RCA-WM / ACWMI and the World-Model-First objects define what a compiled information set is. Empirically, on a real multi-band PIT archive, thick worlds dominate thin worlds and durable bands have large leave-one-out economic value. Selective ACWMI gating is implementable under frozen thresholds but is not CE-dominant versus ungated thick signals here. EvoQuant remains the laboratory, not the theory source.

Appendix. Reproducibility

```
Bootstrap          pdf/sci/bootstrap_multiband_archive.py;          PIT
pdf/sci/build_pit_archive.py; empirics pdf/sci/run_pit_jf_experiments.py; panel
pdf/data/pit_multiband_panel.csv.
```

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